Roll No. 1713450005

SET/SH/BT/C–102

B. Tech. (First Semester)
EXAMINATION, 2017-18

(New Course)

PHYSICS

Time : Two Hours] [*Maximum Marks : 70*

Note : (i) Attempt any *five* questions from Section A and any *three* questions from Section B.

(ii) Answer each question of Section A within 50 words.

(iii) Limit your answers within the given answer book. Additional answer book (B-Answer book) should not be provided or used.

Section—A

Note : Attempt any *five* questions. Each question carries 5 marks.

1. Two straight narrow parallel slits, 2.0 cm apart, are illuminated with a monochromatic light of wavelength 5896^0 A. Fringes are observed at a distance of 60 cm from the slits. Find the width of fringes.

2. Explain population inversion.

P. T. O.

3. What is Curie Weiss law ? Discuss its application for ferromagnetic materials.
4. Explain the basic cause of paramagnetism.
5. Explain Meisner Effect.
6. Define penetration depth for a superconductor. What is its value at critical temperature ?
7. Define specific rotation.
8. Prove that electromagnetic waves are transverse in nature.

Section—B

Note : Attempt any *three* questions. Each question carries 15 marks.

1. Describe and explain the formation of Newton's rings in reflected light. Prove that in reflected light the diameters of bright rings are proportional to the square root of odd numbers.
2. How is the resolving power of a telescope is determined, using a rectangular slit ? Deduce the formula used.
3. Write Maxwell's equation in free space. State the laws on which these equations are based.
4. Describe the construction and working of a Nicol's prism.
5. A parallel beam of sodium light is normally incident on a plane transmission grating, having 5000 lines per centimeter and a second order spectral line is observed at an angle of 30°. Calculate the wavelength of light.
6. What are Einstein's A and B coefficients ? Find a relation between them.

Roll No. 1813450004

SET/SH/BT/C–102

B. Tech. (First Semester)
EXAMINATION, 2018-19

(New Course)

PHYSICS

Time : Two Hours] [*Maximum Marks : 70*

Note : (i) Attempt any *five* questions from Section A and any *three* questions from Section B.

(ii) Answer each question of Section A within 50 words.

(iii) Limit your answers within the given answer book. Additional answer book (B-Answer book) should not be provided or used.

Section—A

Note : Attempt any *five* questions. Each question carries 5 marks.

1. Explain how Nicol's prism can be used as a polariser and as an analyser.
2. Derive an expression for the resolving power of a microscope.

3. Why four-level laser is advantageous than three-level laser ? Explain with example.
4. A thin soap film ($\mu = 1.33$) seen by sodium light ($\lambda = 5893$ Å) by normal reflection appears dark. Find the minimum thickness of the film.
5. Calculate the interplanar spacing for a (321) plane in a simple cubic lattice whose lattice constant is 4.2×10^{-10} m.
6. Explain different defects present in a solid and their effects on properties of that solid.
7. Explain Ferromagnetism and anti-ferromagnetism.
8. What is meant by displacement current and its use in formation of Maxwell's equations ?

Section—B

Note : Attempt any *three* questions. Each question carries 15 marks.

1. Describe and explain the formation of Newton's rings in reflected monochromatic light. Prove that in reflected light (i) diameters of bright rings are proportional to the square-roots of odd numbers and (ii) diameter of dark rings are proportional to the square-roots of natural numbers.
2. Give the construction and theory of a plane diffraction grating of the transmission type and explain the formation of spectra by it.
3. Explain the terms 'absorption', 'spontaneous emission' and 'stimulated emission' of radiation. Obtain a relation between transition probabilities of spontaneous and stimulated emissions.

4. What is meant by intrinsic and extrinsic semiconductors ? Show that Fermi level lies at the centre of the forbidden gap for the intrinsic semiconductor.
5. Explain superconductivity and its types. Give a theory which can explain superconductivity.
6. Obtain the Poynting's theorem for the conservation of energy in an electromagnetic field. Discuss significance of Poynting's vector.

Roll No. 2113405013

P–417

SET/SH/BT/C–102

B. Tech. (First Semester)
EXAMINATION, 2021-22

(New Course)

PHYSICS

Time : Two Hours] [*Maximum Marks : 70*

Note : (i) Attempt any *five* questions from Section A and any *three* questions from Section B.

(ii) Answer each question of Section A within 50 words.

(iii) Limit your answers within the given answer book. Additional answer book (B-Answer book) should not be provided for use.

Section—A

Note : Attempt any *five* questions. Each question carries 5 marks.

1. Two identical coherent waves produce interference pattern. Find the ratio of intensity at the centre of a

bright fringe to the intensity at a point one quarter of the distance between two fringes from the centre.

2. What is meant by diffraction of light ? Distinguish between Fresnel and Fraunhofer class of diffraction.

3. A sugar solution in a tube of length 20 cm produces an optical rotation of 13°. The solution is diluted to one-fourth of its previous concentration. Find the optical rotation produced by 30 cm long tube containing the dilute solution.

4. What do you mean by optical pumping ? Explain population inversion.

5. A photon recoils back after striking an electron at rest. What is the change in the wavelength of the photon ?

6. Define lattice, basis and unit cell.

7. Explain Brillouin zone and effective mass.

8. Explain Ampere's law and displacement current.

Section—B

Note : Attempt any *three* questions. Each question carries 15 marks.

1. What is interference of light ? Calculate the thickness of a thin sheet of transparent material placed in the path of one the interfering beams in double slit experiment.

2. What is meant by double refraction ? Explain uniaxial and biaxial crystals, also write the construction, working and use of Nicol prism.

3. What is Laser ? Explain its principle and characteristics. Also find the relation between Einstein's A and B coefficients.

4. Define Miller indices. Show that for cubic lattice the relation between interplaner distance and Miller indices of a family of planes (h, k, l) is given by :

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

5. Discuss the origin of energy bands in solids on the basis of Kronig-Penny model.

6. Write Maxwell's equations in integral and differential forms. Prove that light is an electromagnetic wave and show that the electric and magnetic vectors are mutually perpendicular to each other in it.

90

Roll No. 151345623016

SET/SH/BT/C–101/ MA–103/BS–101

B. Tech. (First Semester) EXAMINATION, 2015-16

MATHEMATICS—I

Time : Three Hours] [*Maximum Marks : 100*

Note : Attempt total *five* questions selecting at least *one* question from each Section. All questions carry equal marks.

Section—A

1. (a) Show that the transpose of product is the product of the transposes taken in the reverse order.

   (b) Reduce the matrix :

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

   to normal form and then find its rank.

2. (a) Show that every square matrix A can be uniquely expressed as P + i Q, where P and Q are Hermitian matrices.

(b) Show that the equations :

$$5x + 3y + 7z = 4$$
$$3x + 26y + 2z = 9$$
$$7x + 2y + 10z = 5$$

are consistent and solve them.

3. (a) Find the characteristic equation of the matrix :

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 2 & -3 & 0 \\ 1 & 1 & -1 \end{bmatrix}$$

and verify that it is satisfied by A and hence obtain A^{-1}.

(b) Determine the eigen values and corresponding eigen vectors of :

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ 2 & -2 & 0 \end{bmatrix}$$

Section—B

4. Prove that :

(i) $\text{div}\left(\vec{A} \times \vec{B}\right) = \vec{B} \cdot \text{curl}\ \vec{A} - \vec{A} \cdot \text{curl}\ \vec{B}$

(ii) $\text{grad}\left(\vec{A} \cdot \vec{B}\right) = \left(\vec{B} \cdot \nabla\right)\vec{A} + \left(\vec{A} \cdot \nabla\right)\vec{B}$

$$+ \vec{B} \times \text{curl}\ \vec{A} + \vec{A} \times \text{curl}\ \vec{B}$$

5. (a) Verify divergence theorem for :

$$F = \left(x^2 - yz\right)i + \left(y^2 - zx\right)j + \left(z^2 - xy\right)k$$

taken over the rectangular parallelopiped :

$$0 \le x \le a$$
$$0 \le y \le b$$
$$0 \le z \le c$$

(b) Verify Stokes' theorem for $F = \left(x^2 + y^2\right)i - 2xyj$ taken round the rectangle bounded by $x = \pm a, y = 0, y = b$.

Section—C

6. (a) If $y = e^{\tan^{-1} x}$, then show that :

$$\left(1 + x^2\right)y_2 + \left(2x - 1\right)y_1 = 0$$

and

$$\left(1 + x^2\right)y_{n+2} - \left\{2\left(n + 1\right)x - 1\right\}y_{n+1} + n\left(n + 1\right)y_n = 0$$

(b) Prove that :

$$\log_e\left(x + h\right) = \log_e h + \frac{x}{h} - \frac{x^2}{2h^2} + \frac{x^3}{3h^3} - \ldots\ldots\ldots$$

7. (a) If :

$$z = xyf\left(\frac{y}{x}\right)$$

show that :

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 2z$$

and if z is a constant, then show that :

$$\frac{f'\left(\frac{y}{x}\right)}{f\left(\frac{y}{x}\right)} = \frac{x\left(y + x\frac{dy}{dx}\right)}{y\left(y - x\frac{dy}{dx}\right)}$$

(b) Find the envelope of the family of the curves :

$$\sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} = 1$$

when $a^n + b^n = c^n$, c bring any constant.

8. (a) Find the maximum or minimum values of $u = x^3y^2(1 - x - y)$.

(b) Trace the curve :

$$y^2(a + x) = (a - x)x^2$$

Section—D

9. (a) Discuss Binomial distribution. Also find the constants of the Binomial distribution.

(b) Show that for the Poisson distribution with mean m, :

$$\mu_{r+1} = rm\mu_{r-1} + m\frac{d\mu_r}{dm}$$

10. Find the correlation coefficient and the equations of regression lines for the following values of x and y :

x	y
1	2
2	5
3	3
4	8
5	7

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SET/SH/BT/C–101

B. Tech. (First Semester)
EXAMINATION, 2018-19

MATHEMATICS—I

Time : Two Hours] [*Maximum Marks : 70*

Note : (i) Attempt any *five* questions from Section A and any *three* questions from Section B.

(ii) Answer each question of Section A within 50 words.

(iii) Limit your answers within the given answer book. Additional answer book (B-Answer book) should not be provided or used.

Section—A

Note : Attempt any *five* questions. Each question carries 5 marks.

1. If $\vec{f} = i(x + y + 1) + j + k(-x - y)$, then prove that :

$$\vec{f} \cdot \text{curl } \vec{f} = 0$$

2. If A is a symmetric matrix, then prove that adj A is also symmetric.

3. Find the rank of the following matrix :

$$A = \begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 9 \\ -1 & -3 & -4 & -3 \end{bmatrix}$$

4. If $u = \sin^{-1}\left(\frac{x^2 + y^2}{x + y}\right)$, show that :

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u$$

5. Find the asymptotes of the following curve :

$$y^3 = x^3 + ax^2$$

6. Discuss the maxima and minima of the function :

$$u = x^3 + y^3 - 3axy$$

7. Define normal distribution with example.

Section—B

Note : Attempt any *three* questions. Each question carries 15 marks.

8. Show that :

$$\int_S (axi + byj + czk).\hat{n}\, dS = \frac{4}{3}\pi(a + b + c)$$

where S is the surface of the sphere $x^2 + y^2 + z^2 = 1$.

9. Show that the equations :

$$x + y + z = 6$$
$$x + 2y + 3z = 14$$
$$x + 4y + 7z = 30$$

are consistent and solve them.

10. Define the following :
    (a) Cylindrical and spherical co-ordinate system
    (b) Leibnitz theorem
    (c) Probability

11. Trace the following curve :

$$y^2(a + x) = x^2(3a - x)$$

12. Find the coefficient of correlation for the following data :

x	y
1	8
3	12
5	15
7	17
8	18
10	20

13. (a) Obtain the mean and first three central moments of the Poisson distribution from the binomial distribution.
    (b) Expand sin x with the help of Maclaurin's theorem.

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P–416
SET/SH/BT/C–101

B. Tech. (First Semester)
EXAMINATION, 2021-22
MATHEMATICS—I

Time : Two Hours] [*Maximum Marks : 70*

Note : (i) Attempt any *five* questions from Section A and any *three* questions from Section B.

(ii) Answer each question of Section A within 50 words.

(iii) Limit your answers within the given answer book. Additional answer book (B-Answer book) should not be provided for use.

Section—A

Note : Attempt any *five* questions. Each question carries 5 marks.

1. If :

$$\vec{F} = xy^2\hat{i} + 2x^2yz\hat{j} - 3yz\hat{k},$$

then find div $(\vec{F})$ at the point $(1, -1, 1)$.

(6)

P. T. O.

2. Evaluate the curl of the vector :

$$V = (x^2 + yz)\hat{i} + (y^2 + zx)\hat{j} + z^2\hat{k}$$

at the point (1, 2, 3).

3. If :

$$u = \log\left(\frac{x^2 + y^2}{x + y}\right),$$

prove that :

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 1.$$

4. If :

$$x = r\cos\theta,$$

$$y = r\sin\theta$$

prove that :

$$\frac{\partial(x, y,)}{\partial(r, \theta)} = r.$$

5. Find the asymptotes of the curve :

$$a^2y^2 - b^2x^2 = x^2y^2$$

6. Discuss the Poisson's distribution.

7. Find the rank of the matrix :

$$A = \begin{bmatrix} 5 & 3 & 0 \\ 1 & 2 & -4 \\ -2 & -4 & 8 \end{bmatrix}$$

Section—B

Note : Attempt any *three* questions. Each question carries 15 marks.

8. (a) Prove that :

$$\text{div}(A + B) = \text{div}(A) + \text{div}(B)$$

(b) Evaluate :

$$\int_S \vec{F} \cdot n\, dS,$$

where $\vec{F} = ax\hat{i} + by\hat{j} + czk$ and S is the surface of sphere $x^2 + y^2 + z^2 = 1$.

9. Verify Stokes' theorem when $\vec{F} = x^2\hat{i} + xy\hat{j}$, where C is the perimeter of the square in XY-plane whose sides are along the lines $x = 0, y = 0, x = a, y = a$.

10. (a) Find the solution of the given system of linear equations using Cramer's rule :

$$x + y - z = 6$$
$$3x - 2y + z = -5$$
$$x + 3y - 2z = 14.$$

(b) Find the characteristic equation for the matrix :

$$A = \begin{bmatrix} 3 & 6 \\ 5 & -4 \end{bmatrix}$$

11. (a) Find all eigenvalues and corresponding eigenvectors for the matrix :

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

(b) Verify Caley-Hamilton theorem for the square matrix :

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}$$

12. (a) Expand $\log(1+x)$ by Maclaurin's theorem.

(b) If u be a homogeneous function of x and y of degree n, then :

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = nu .$$

13. (a) Find the evalote of the parabola $y^2 = 4ax$.

(b) Trace the curve :

$$ay^2 = x^3 .$$

Roll No. 1813450008

SET/CS/BT/C–103

B. Tech. (CS/IT) (First Semester) EXAMINATION, 2018-19

(New Course)

CHEMISTRY

Time : Two Hours] [*Maximum Marks : 70*

Note : (i) Attempt any *five* questions from Section A and any *three* questions from Section B.

(ii) Answer each question of Section A within 50 words.

(iii) Limit your answers within the given answer book. Additional answer book (B-Answer book) should not be provided or used.

Section—A

Note : Attempt any *five* questions. Each question carries 5 marks.

1. Explain activation energy of catalysis.

2. State first law of thermodynamics. Write its application and limitation.

3. How would you differentiate between molecularity and order of reaction ?
4. Discuss EDTA method for determining the hardness of water.
5. Discuss characteristics of good fuels.
6. Write short notes on the following :
   (a) Catalytic promoters
   (b) Acid rain
7. Write a short note on Synthesis rubbers.

Section—B

Note : Attempt any *three* questions. Each question carries 15 marks.

1. Complete the following reactions. Give their names and explain their mechanism :

   (a) $2\,H-\overset{\overset{O}{\|}}{C}-H + NaOH \xrightarrow{\Delta}$

   (b) $\begin{matrix} C_6H_5 \\ H_3C \end{matrix}\!\!>\!C=N-OH \xrightarrow{\text{Conc. } H_2SO_4}$

2. What do you understand by corrosion of metals ? Give its causes and explain electrochemical theory of corrosion.
3. (a) Explain Octane and Cetane numbers of fuels.
   (b) Write a note on mechanism of Lubrication.

4. What are the characteristics of catalytic reactions ? Discuss briefly the adsorption theory of catalysis.
5. How does the valency bond theory account for the following facts ?
   (i) $\left[Ni(CN)_4\right]^{2-}$ is diamagnetic and square planar.
   (ii) $[NiCl_4]^{2-}$ is paramagnetic and tetrahedral.
   (iii) $\left[Ni(CO)_4\right]$ is diamagnetic and tetrahedral.

Roll No. .....

P–418

SET/CS/BT/C–203

B. Tech. (CS/IT) (First Semester) EXAMINATION, 2021-22

(New Course)

CHEMISTRY

Time : Two Hours] [*Maximum Marks : 70*

Note : (i) Attempt any *five* questions from Section A and any *three* questions from Section B.

(ii) Answer each question of Section A within 50 words.

(iii) Limit your answers within the given answer book. Additional answer book (B-Answer book) should not be provided for use.

Section—A

Note : Attempt any *five* questions. Each question carries 5 marks.

1. State and explain first law of thermodynamics. Derive the work done for reversible isothermal expansion of ideal gas.

2. Explain inner and outer orbital octahedral complexes of transition elements with suitable examples.
3. What are conducting polymers ? Discuss their applications in brief.
4. Discuss the preparation of *two* synthetic rubbers.
5. Discuss the properties of lubricants.
6. Explain Galvanization and tinning methods for protection from corrosion.
7. What is Biogas ? Discuss the main steps involved in the formation of biogas.

Section—B

Note : Attempt any *three* questions. Each question carries 15 marks.

1. Discuss the mechanism of the following reactions :
   (a) Reimer-Tiemann Reaction
   (b) Cannizzaro Reaction
   (c) Diels-Alder Reaction
2. Discuss the main points of crystal field theory. Show the splitting of *d*-orbitals in octahedral complexes. What are the various factors which affect the magnitude of crystal field splitting energy ?

3. Explain the following terms :
   (a) Hardness of water
   (b) Joule-Thomson Effect
   (c) Depletion of ozone layer
4. What are the boiler problems in industries caused by hard water ? Discuss the method to overcome these boiler problems. Explain lime soda process for purification of water.
5. What is calorific value of a fuel ? Discuss the working of bomb colarimeter to determine calorific value of fuels.

Roll No. 1613450300 1

SET/CS/BT/C–105

B. Tech. (CS/IT) (First Semester) EXAMINATION, 2016-17

(New Course)

COMPUTER PROGRAMMING

Time : Two Hours] [*Maximum Marks : 70*

Note : (i) Attempt any *five* questions (out of seven questions) from Section A and any *three* questions (out of six questions) from Section B.

(ii) Answer each question of Section A within 50 words.

(iii) Limit your answers within the given answer book. Additional answer book (B-Answer book) should not be provided or used.

Section—A

Note : Attempt any *five* questions. Each question carries 5 marks.

1. With the help of suitable examples, describe :
   (a) Logical operators
   (b) Conditional operators

2. What are Library Functions ? With the help of examples, discuss the utilities of library functions.

3. With the help of suitable examples, discuss the following :
   (a) Static variables
   (b) Register variables
   (c) External variables
   (d) Auto variables

4. Write a 'C' program for generating a series :
   100, 98, 96, 94, ..., 0
   Find the sum of series also.

5. What do you understand by recursion ? Write down a 'C' program for finding out HCF (Highest Common Factor) of two given numbers, with the help of recursion.

6. Write down the multidimensional arrays' advantages. How are they declared in 'C' language ? If the base address of array :
   $x[10][40]$ is 10000
   what will be the address of :
   (a) $x[1][5]$
   (b) $x[5][39]$
   (c) $x[0][0]$
   (d) $x[2][10]$

7. What do you mean by dynamic memory allocation ? Explain with the help of 'C' library functions.

Section—B

Note : Attempt any *three* questions. Each question carries 15 marks.

1. Write down a 'C' program that will read names of source and destination files and copy the contents of source file into destination file.
2. What are self-referential structures ? Declare a self-referential structure with the help of 'C' language. Write down the applications of self-refential structures.
3. Write down a 'C' program for swaping the contents of two variables with the help of :
   (a) Call by value
   (b) Call by reference
4. What is a function in 'C' ? Write a functions in 'C' that will find the maximum number among the given 'n' numbers, were $n \geq 100$.
5. Differentiate between the following :
   (i) Pointer and Arrays
   (ii) ! and ! = operators
   (iii) 'break' and 'continue' statement
   (iv) 'float' and 'double' data type
6. Write short notes on the following :
   (a) Keywords in 'C'
   (b) Array of structures
   (c) Global variables
   (d) 'Go to' statement
   (e) 'do while' loop

Roll No. 1.B.124.501.026

SET/ME/BT/C–102

B. Tech. (First Semester)
EXAMINATION, 2018-19

BASIC MECHANICAL ENGINEERING

Time : Two Hours] [*Maximum Marks : 70*

Section—A

Note : Attempt any *five* questions. Each question carries 5 marks.

1. Prove that the difference between two specific heat is equal to characteristic gas constant.
2. A mass of 0.25 kg of air in a closed system expands from 2 bar, 60°C to 1 bar and 40°C while receiving 1.005 kJ of heat from a reservoir at 100°C. The surrounding atmosphere is at 0.95 bar and 27°C. Determine the maximum work. How much of this work would be done on the atmosphere ?
3. Explain the adiabatic process. Derive an expression for the work done during the adiabatic expansion of an ideal gas.
4. A hollow cast iron cylinder 4 m long, 300 mm outer diameter and thickness of metal 50 mm is subjected to a central load on the top when standing straight. The

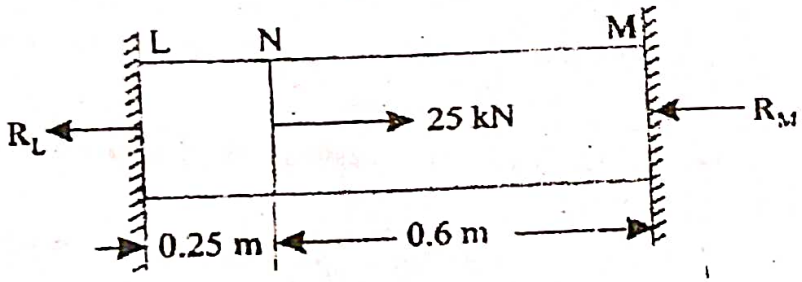
stress produced 75000 kN/m^2. Assume Young's Modulus for cast iron is 1.5×10^8 kN/m^2. Find :

(i) Magnitude of the load

(ii) Total decrease in length

5. Write a short note on Capstan and Turret lathe.

6. Figure shows a steel bar of area 250 mm^2 held firmly by the end supports and loaded by and axial load of 25 kN. Determine (i) reaction at L and M (ii) extension of the left portion. If E = 200 GN/m^2.



7. Derive the relation between Young modulus, Modulus of rigidity and Bulk modulus.

Section—B

Note : Attempt any *three* questions. Each question carries 15 marks.

1. (a) Describe with a neat sketch, water level indicator for a boiler. Enlist the various heat losses in boiler. Which is the biggest loss and why ? 10

   (b) Define Enthalpy. Why the enthalpy of an ideal gas does depends only on the temperature ? 5

2. A volume of 0.14 m^3 of air at 1 bar and 90°C is compressed to 0.0 14 m^3 according to $PV^{1.3}$ = constant.

Heat is then added at a constant volume until the pressure is 66 bar. Determine : 15

(i) heat exchange with cylinder walls during compression

(ii) change of entropy during each portion of the process.

Adiabatic index =1.4, R= 286 J/kg-K.

3. (a) Explain the working principle of lathe machine with neat sketch. Also discuss the different operation on lathe machine. 10

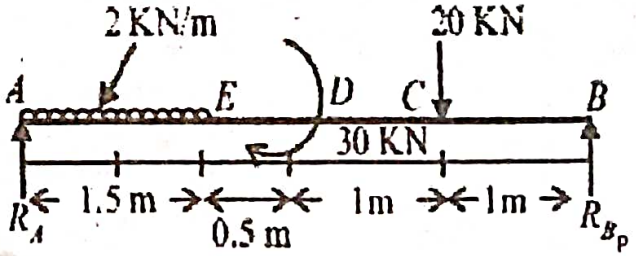
   (b) Define the following : 5

   (i) Macroscopic and Microscopic

   (ii) Intensive and Extensive

   (iii) Closed system and Open system

4. Draw the SFD and BMD for the simply supported beam loaded as shown in Figure. 15



5. (a) Draw the Mohr stress circle for direct stresses of 65 MN/m^2 (tensile) and 35 . 65 MN/m^2

(compressive) and estimate the magnitude and direction of the resultant stresses on the planes making angle of 20°C and 65°C with the plane of the first principal stress. Find also the tangential stresses on these planes. 8

(b) Draw Stress strain diagram for Cast iron and Mild Steel and explain their salient features. 7

Basic Mechanical Engineering Time=1hr

Q1. Draw stress strain diagram for ductile and brittle materials mark and explain their salient points and also give two examples of each.

A steel bar of 250mm long, its two ends are 40mm and 30 mm in diameter and the length of each rod is 200mm. the middle portion of the bar is 15mm in diameter and 500mm long. If the bar is subjected to an axial tensile load of 15KN, find its total extension and also determine the stress produce in each portion.

Q2 Derive Torque Equation.

A point in a strained material is subjected to a tensile stress of 65 N/mm^2 and a compressive stress of 45 N/mm^2, acting on two mutually perpendicular planes and a shear stress of 10 N/mm^2 are acting on these planes. Find the normal stress, tangential stress and resultant stress on a plane inclined to 30° with the plane of the compressive stress.

Roll No. ...1813.45.0.10.26

SET/ME/BT/C–104

B. Tech. (First Semester)

EXAMINATION, 2018-19

ENGINEERING MECHANICS

Time : Two Hours] [*Maximum Marks : 70*

Note : (i) Attempt any *five* questions from Section A and any *three* questions from Section B.

(ii) Answer each question of Section A within 50 words.

(iii) Limit your answers within the given answer book. Additional answer book (B-Answer book) should not be provided or used.

Section—A

Note : Attempt any *five* questions. Each question carries 5 marks.

1. Explain the following terms :

   (i) Law of transmissibility

   (ii) Rigid Body

   (iii) Structure

2. Determine the centroid of semicircle.

3. Explain the types of frame.

4. A man of weight W = 712 N holds one end of a rope that passes over a pulley vertically above his head and to the other end of which is attached a weight Q = 534 N. Find the force with which the man's feet press against the floor.

5. If a stone falls passed a window of 2.45 m height in half a second, find the height from which stone is felt.

6. A projectile is aimed at a target on the horizontal plane and falls 12 m short when the angle of projection is 15°, while it overshoots by 24 m when the angle is 45°. Find the angle of projection to hit the target.

7. State and prove Lami's theorem.

8. A body weighing 300 N is pushed up a 30° plane by a 400 N force acting parallel to the plane. If the initial velocity of the body is 1.5 m/s and coefficient of kinetic friction is $\mu = 0.2$, what velocity will the body have after moving 6 m ?

Section—B

Note : Attempt any *three* questions. Each question carries 15 marks.

1. Two cars are travelling towards each other on a single lane road at the velocities 12 m/s and 9 m/s respectively. When 100 m apart, both drivers realize the situation and apply their brakes. They succeed in stopping simultaneously and just short of colliding. Assume constant deceleration for each case and determine :

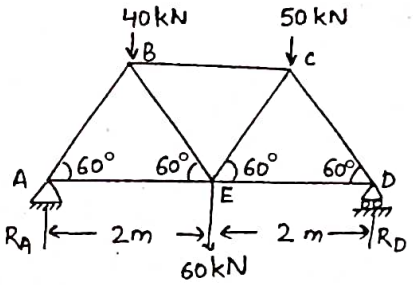
   (a) time required to stop the cars

   (b) deceleration of each car and the distance travelled by each car while slowing down.

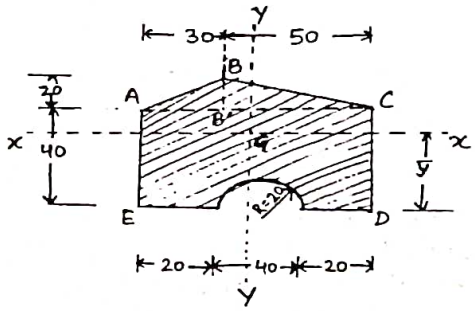
2. (a) Determine the time of flight, range and the greatest height reached by the projection on level ground.

   (b) Derive an expression for mass moment of inertia of sphere about its axis of rotation of diameter D.

3. Determine all the forces acting on members of truss. Indicate with magnitude and direction in diagram.



4. Find the second moment of the shaded portion about the central axis.



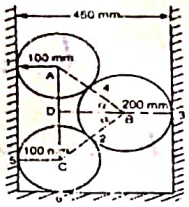
5. Explain the following :

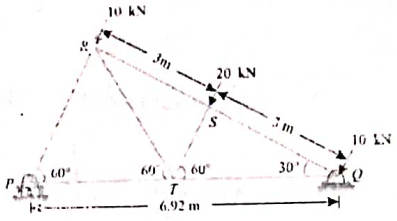
   (i) Various types of beams

   (ii) Parallel axis and Perpendicular axis theorem.

ENGINEERING MECHANICS

1. The spheres A, B and C weighing 80 N, 160 N and 80 N respectively are having radii of 100 mm, 200 mm and 100 mm respectively are placed in a trench as shown in Fig 1. Treating all contact surfaces as smooth, determine the reactions developed. **06 Marks**
2. Explain the following: (i) Principle of transmissibility of forces (ii) Perfect, Deficient & Redundant frame. **03 Marks**
3. Determine the forces in all the members of truss as shown in Fig. 2. **06 Marks**





B.Tech. (First Semester) Examination, 2016

MECHANICAL ENGINEERING

Time: 1 1/2 Hours] [*Maximum Marks: 30*

Note: Attempt all questions.

1. (a) Define the following:
   (i) Specific volume
   (ii) Thermodynamic system
   (iii) Intensive and extensive property
   (iv) Zeroth law
   (v) Heat engine, heat pump and refrigerator

   (b) A volume of 0.14 m^3 of air at 1 bar and 90°C is compressed to 0.014m^3 according to the law $PV^{1.3}$= C. Heat is then added at constant volume till the pressure is 50 bar. Determine (i) Heat exchange with cylinder walls during compression. (−3.55 kJ)

   (c) 15 kg of air per minute is delivered by a centrifugal air compressor. The inlet and outlet conditions of air are V_1 = 10 m/s, p_1 = 1 bar, vs_1 0.5 m^3/kg and V_2 = 80 m/s, p_2 = 7 bar, vs_2=0.15 m^3/kg. The increase in enthalpy of air passing through the compressor is 160 kJ/kg, and heat lost to the surroundings is 720 kJ/min. Assuming that inlet and discharge are at the same level, find:
   (i) motor power required to drive the compressor,
   (ii) ratio of inlet and outlet pipe diameter.

OR

2. (a) "No engine can be more efficient than a reversible engine" Justify this statement.
   (b) Prove equivalence of Kelvin Planck and Clausius statement.
   (c) A reversed Carnot cycle operates at either a refrigerator or heat pump. In either case, the power input is 20.8 kW. Calculate the quantity of heat extracted from the cold body for either type of machine. In both case 3500 kJ/min heat is delivered by the machine. In case of the refrigerator the heat is transferred to the surroundings while in case of heat pump, the space is to be heated. What is their respective coefficient of performances? If the temperature of cold body is 0°C for the refrigerator and 5°C for heat pump what will be respective temperatures of surrounding for refrigerator and heated space for heat pump? What reduction in heat rejection temperatures would be achieved by doubling the COP for same cold body temperature?

Ritik Kumar

B.Tech (Class Test)

Engineering Mechanics

Max. Marks: 15 Duration: 1 hr

Note: Attempt any three questions. Every question consists of equal marks.

Question 1. Explain law of mechanics.

Question 2. Two smooth circular cylinders each of weight W = 445 N and radius r = 152 mm are connected at their centres by a string AB of length l = 406 mm and rest upon a horizontal plane, supporting above them a third cylinder of weight Q = 890 N and radius r = 152 mm. Find the forces in the string and the pressures produced on the floor at the point of contact.

Question 3. A man of weight W = 712 N holds one end of a rope that passes over a pulley vertically above his head and to the other end of which is attached a weight Q = 534 N. Find the force with which the man's feet press against the floor.

Question 4.Find out the centroid of a triangle.

Roll No. 1713450025

SET/SH/BT/C–104

B. Tech. (First Semester)
EXAMINATION, 2017-18

BASIC ELECTRONICS

Time : Two Hours] [*Maximum Marks : 70*

Note : (i) Attempt any *five* questions from Section A and any *three* questions from Section B.

(ii) Answer each question of Section A within 50 words.

(iii) Limit your answers within the given Answer Book. Additional Answer Book (B-Answer Book) should not be provided or used.

Section—A 5 each

1. Explain the working and construction of PN junction diode.
2. Explain the working of full wave bridge rectifier circuit.
3. Explain the working, construction and characteristics of NPN transistor.
4. How can a BJT used as a switch ?
5. List the differences between BJT and FET.
6. Explain the construction and working of enhancement type *n*-channel MOSFET.

7. Explain the characteristics of an Ideal OP-AMP.

Section—B

8. (a) Convert the following : 2 each

(i) $(31)_{10} = (?)_2\ (?)_4 = (?)_8 = (?)_{16}$

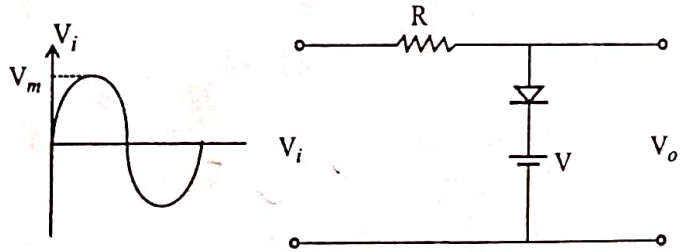
(ii) $(11001)_2 = (?)_{Gray}$

(iii) Calculate the 2's complement of $(1101011)_2$

(b) Explain the implementation of NAND and NOR gate using *n*-channel MOSFET. 9

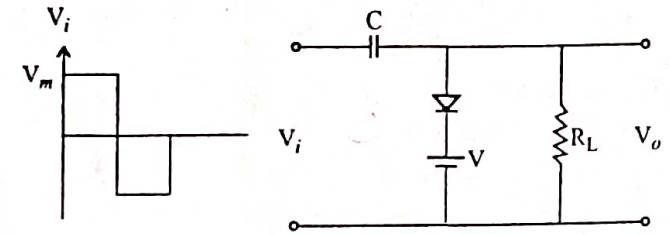
9. (a) Draw the O/P waveform for the given circuits. 4 each

(i)



where $V_m > V$.

(ii)



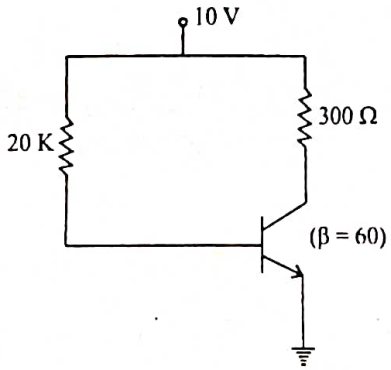
where $V_m > V$.

(b) Explain fixed biasing configuration circuit. For the given circuit. Determine :

(i) The collector current I_C

(ii) Collector emitter voltage V_{CE}

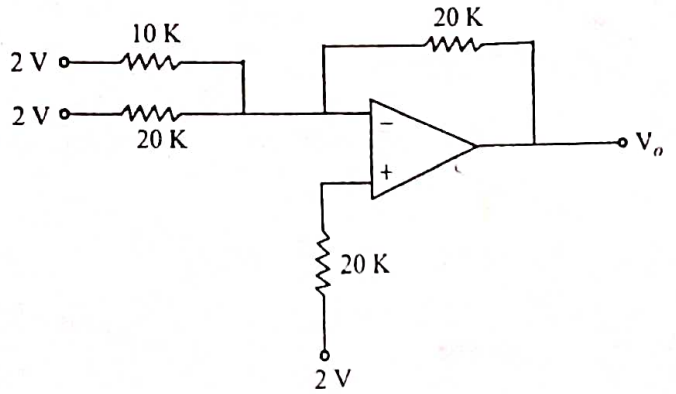
(Assume $V_{BE} = 0.6$ V and $\beta = 60$) 7



10. Implement AND, OR, NOT and NOR gates using NAND gates. 15

11. (a) Draw the circuit of an OP-AMP integrator and explain its operation. 7

(b) Calculate the output voltage (V_o) for the given ideal OP-AMP circuit. 8



12. Explain the operation of CMOS type NAND and NOR gates circuit with the help of neat circuit diagram. 15

13. Write the short notes on any *three* of the following : 5 each

(i) Intrinsic and extrinsic material

(ii) Operating regions of BJT

(iii) Fundamental theorems of Boolean Algebra

(iv) Representation of signed number in binary

(v) OP-AMP as a summer amplifier.

Roll No. 1.8.1.3.9.5.0.9.0.0.6

SET/SH/BT/C–104

B. Tech. (First Semester) EXAMINATION, 2018-19

BASIC ELECTRONICS

Time : Two Hours] [*Maximum Marks : 70*

Note : (i) Attempt any *five* questions from Section A and any *three* questions from Section B.

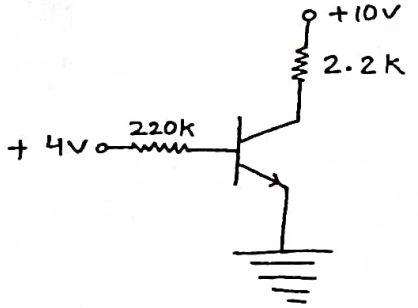
(ii) Answer each question of Section A within 50 words.

(iii) Limit your answers within the given answer book. Additional answer book (B-Answer book) should not be provided or used.

Section—A 5 each

1. Draw and explain the ideal diode characteristics.
2. Explain working, construction and characteristics of a NPN transistor.
3. Explain the working of full wave bridge rectifier circuit. Also determine the V_{rms}, V_{av}.
4. Explain the term forbidden gap. Draw energy band diagram for insulator, semiconductor and conductor.
5. Describe the *n*-channel enhancement type MOSFET and its characteristics.

6. Calculate the terminal voltages (V_B, V_C, V_E) and terminal currents (I_B, I_C, I_E) for the given circuit in Figure. 1. Given $V_{BE} = 0.7$ V and $\beta = 200$.



7. Convert the following :

   (i) $(283)_9 = (?)_{16} = (?)_8 = (?)_4 = (?)_2$

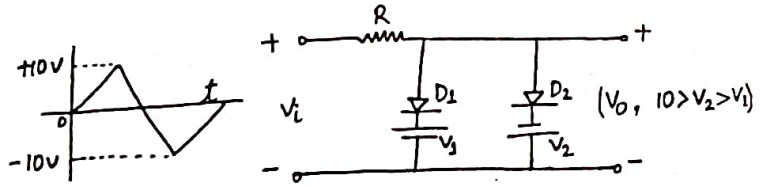
   (ii) $(1011011)_2 = (?)_{Gray}$

   (iii) $(211)_x = (152)_8$, find x.

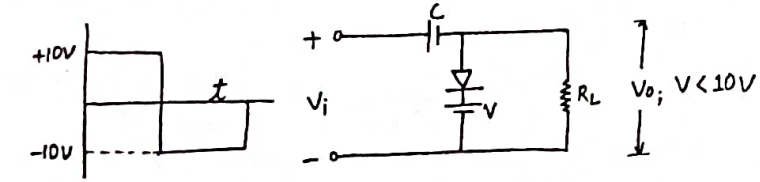
Section—B

8. Sketch the output waveform for given circuits, for their input wave (consider all diode as ideal diode) : 8, 7

(i)



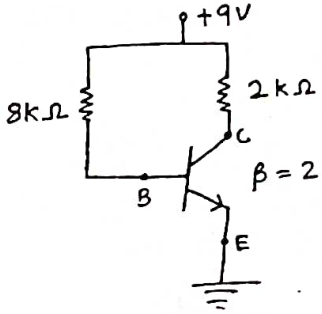
(ii)



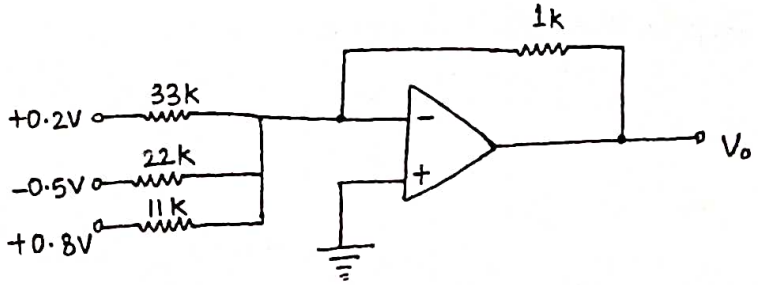
9. Explain all types of methods, which are used for obtaining transistor biasing. 15

10. (a) Draw NAND and NOR gate circuit with the help of CMOS logic diagram/design. 7

    (b) Draw the d.c. load line and locate the operating point for the fixed biasing transistor shown below (assume $V_{BE} = 0.6$ V). 8



11. (a) Explain the characteristics of ideal Op-amp. Also calculate the output voltage (V_o) of the given circuit : 8



    (b) Explain the operation of integrator circuit of Op-Amp. and derive the expression for output voltage for this circuit. 7

12. (a) Implement all basic gates using : 7

(i) Only NAND gates

(ii) Only NOR gates

(b) $F = AB + \overline{A}\,\overline{B}C + \overline{A}B\overline{C}$ express their function in : 8

(i) Canonical SOP form

(ii) Canonical POS form

13. Write short notes on any *four* of the following : 15

(i) CMRR of Op-amp.

(ii) Intrinsic and extrinsic material

(iii) De-Morgan's theorem of Boolean algebra

(iv) Comparison between BJT and FET

(v) Zener diode and its breakdown

(vi) Operating regions of transistor

Roll No. .....

AECC–104(EVS)

B. B. A./B. H. M./B. Pharm./B. Tech./ B. A. M. C. (First Semester) EXAMINATION, 2018-19

(New Course)

ENVIRONMENTAL SCIENCES

Time : Two Hours] [*Maximum Marks : 70*

नोट : (i) खण्ड 'अ' से किन्हीं **पाँच** प्रश्नों के और खण्ड 'ब' से किन्हीं **तीन** प्रश्नों के उत्तर दीजिए।

Attempt any *five* questions from Section A and any *three* questions from Section B.

(ii) खण्ड 'अ' के प्रत्येक प्रश्न का उत्तर 50 शब्दों तक सीमित रखें।

Answer each question of Section A within 50 words.

(iii) अपने सभी प्रश्नों के उत्तर आपको दी गयी उत्तर पुस्तिका में ही दीजिये। अतिरिक्त उत्तर पुस्तिका नहीं दी जायेगी।

Limit your answers within the given answer book. Additional answer book (B-Answer book) should not be provided or used.

खण्ड—अ

(Section—A)

नोट : किन्हीं पाँच प्रश्नों के उत्तर दीजिए। प्रत्येक प्रश्न 5 अंक का है।

Attempt any *five* questions. Each question carries 5 marks.

1. खाद्य शृंखला क्या है ? उदाहरण सहित समझाइए।

   What is Food Chain ? Describe with examples.

2. सतत् विकास को परिभाषित कीजिए।

   Define sustainable development.

3. नवीकरणीय ऊर्जा किसे कहते हैं ? किन्हीं **तीन** नवीकरणीय ऊर्जा के श्रोतों के नाम लिखिए।

   What do you know by renewable energy ? Name any *three* sources of renewable energy.

4. जैवविविधता के कौन से तीन स्तर हैं ?

   What are the three levels of biodiversity ?

5. हरित गृह प्रभाव से आप क्या समझते हैं ?

   What do you mean by green house effect ?

6. मौन्ट्रीयल प्रोटोकॉल क्या है ?

   What is Montreal Protocol ?

7. चिपको आन्दोलन के बारे में संक्षिप्त विवरण लिखिए।

   Write briefly on Chipko Movement.

खण्ड—ब

(Section—B)

नोट : किन्हीं **तीन** प्रश्नों के उत्तर दीजिए। प्रत्येक प्रश्न 15 अंक का है।

Attempt any *three* questions. Each question carries 15 marks.

1. पारिस्थितिकीय तंत्र क्या है ? किसी जलीय पारिस्थितिकीय तंत्र की संरचना तथा उसके अवयवों की सचित्र व्याख्या कीजिए।

   What is an Ecosystem ? Discuss the structure and its components of any aquatic ecosystem with suitable diagram.

2. जैवविविधता के संरक्षण के विभिन्न तरीकों का वर्णन कीजिए।

   Describe various ways of biodiversity conservation.

3. गंगा के जल प्रदूषण के मुख्य कारणों, प्रभावों तथा नियंत्रण की व्याख्या कीजिए।

   Discuss the major causes, effects and control water pollution of Ganga.

4. ओजोन परत के क्षरण के मुख्य कारकों, प्रक्रिया तथा प्रभाव की विस्तृत व्याख्या कीजिए।

   Describe in detail the major factors, process and impact of ozone layer depletion.

$O + O \rightarrow O_2$

$O_2 + O \rightarrow O_3$

$O_3 + O \rightarrow \frac{1}{2} O_2$

5. ठोस अपशिष्ट क्या है ? ठोस अपशिष्ट प्रबन्ध के विभिन्न तरीकों का वर्णन कीजिए।

What is Solid Waste ? Describe various methods of solid waste management.

6. वन्यजीव संरक्षण अधिनियम, 1972 की प्रमुख विशेषताओं का वर्णन कीजिए।

Describe the salient features of Wildlife Protection Act, 1972.

Roll No. .....

P–104

AECC–104(EVS)

B. B. A./B. H. M./B. Pharm./B. Tech./B. Sc. Yoga/B. A. M. C. (First Semester)

EXAMINATION, 2021-22

(New Course)

ENVIRONMENTAL SCIENCES

Time : Two Hours] [*Maximum Marks : 70*

Note : (i) Attempt any *five* questions from Section A and any *three* questions from Section B.

(ii) Answer each question of Section A within 50 words.

(iii) Limit your answers within the given answer book. Additional answer book (B-Answer book) should not be provided or used.

P. T. O.

Section—A

Note : Attempt any *five* questions. Each question carries 5 marks.

1. What do you mean by Sustainability in Business ?
2. Define an ecosystem.
3. Distinguish between renewable and non-renewable energy with examples.
4. Mention the causes and consequences of deforestation.
5. Explain how our growing energy needs can be met from renewable sources of energy.
6. How far the Kyotoprotocol was successful ?
7. What are the salient features of Chipko Movement ?

Section—B

Note : Attempt any *three* questions. Each question carries 15 marks.

8. Discuss the issues of resettlement and rehabilitation of project affected persons. Suggest some remedial measures for resettlement.
9. Explain the role of religious and cultural values in the protection of environment. What do you mean by environmental ethics ?
10. Discuss the causes and effect of air, water, soil and noise pollution.

11. Elaborate the causes of climate change. Suggest measures to reduce global warming.
12. What are threats of bio-diversity to wild life and bio-diversity ?
13. Explain the disaster management with respect to floods and earthquake.

1750

Roll No. 1813489202

SET/IT/BT/C–105

B. Tech. (First Semester)
EXAMINATION, 2018-19
FUNDAMENTALS OF INFORMATION TECHNOLOGY

Time : Two Hours] [*Maximum Marks : 70*

Note : (i) Attempt any *five* questions from Section A and any *three* questions from Section B.

(ii) Answer each question of Section A within 50 words.

(iii) Limit your answers within the given answer book. Additional answer book (B-Answer book) should not be provided or used.

Section—A

Note : Attempt any *five* questions. Each question carries 5 marks.

1. Differentiate between entry controlled loop and exit controlled loop with syntax.
2. Discuss any *five* major areas where computer can be used or currently being used.
3. (a) Convert Decimal to Binary :

   545.

(b) Convert Hexadecimal to Decimal :

2B1.C4.

4. Discuss any *two* input devices and their working principle in short.

5. Differentiate between the following :

(a) LAN and WAN

(b) Interpreter and Compiler

6. Discuss Ring Network Topology with its advantages and disadvantages.

7. Discuss *five* major areas where internet can be used or currently being used.

Section—B

Note : Attempt any *three* questions. Each question carries 15 marks.

8. Discuss Von-Neumann architecture of computer system along with various components involved in modern day computer system with the help of appropriate diagrams.

9. Write program in C for the following :

(a) Which takes marks of three subjects of a student from user and calculate and print the percentage P and division D on screen (formula for division :

if P > = 60 then D = First, else if 60 > P > = 45 then D = Second, else if 45 > P > = 35 then D = Third, else D = Fail).

(b) Which takes a number from user and print factorial of that number ?

10. How is Information different from Data ? What is an Information System ? Discuss various types of Information Systems available and their use in brief.

11. What is HTML ? Using HTML create a form for registration of students to a university, using radio buttons, check boxes, drop down menus and other controls wherever necessary.

12. Write short notes on the following :

(a) FTP

(b) EEPROM

(c) System Software *vs.* Application Software

(d) Process

(e) e-Governance

